

2.5 Carrier Statistic

2.5.1 Density of States

2.5.2 Fermi Function & Fermi Energy

- Physical Interpretation
- Characteristics

2.5.3 Carrier Densities

- Intrinsic/Extrinsic Semiconductor
- Intrinsic Fermi Energy
- Mass Action Law
- Temperature Dependence

2.5.4 Charge Neutrality Relationship

2.5.5 Non-Complete Ionization

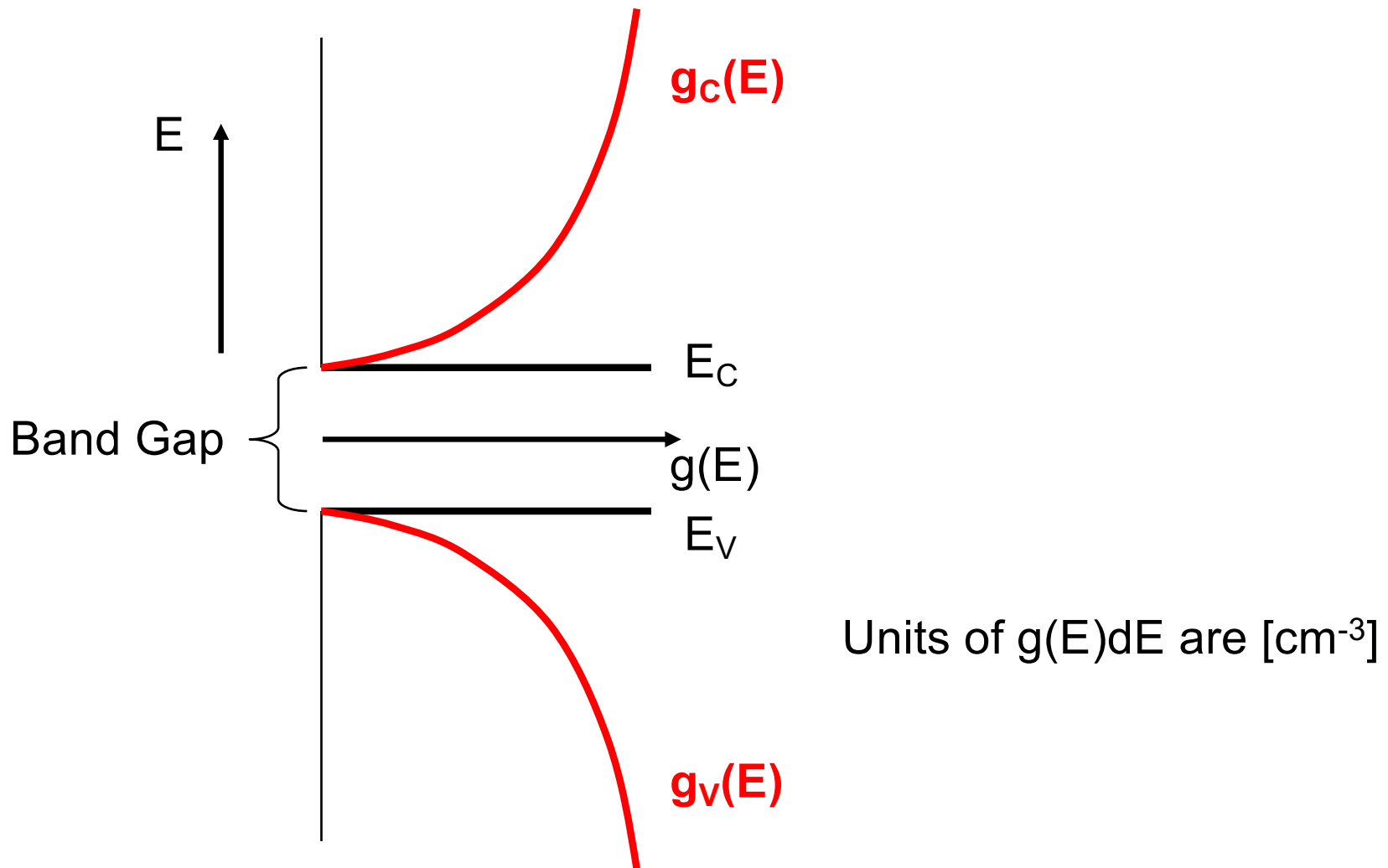
Pierret, Chapter 2.4-2.6, page 40-68

2.5.1 Density of States

From quantum mechanics, we not only obtain the band structure, i.e., the $E(k)$ relations, but also the density of states $g(E)dE$, i.e., how many allowed states are in the range $E \dots E+dE$:

$$g_C(E)dE = \frac{m_n^* \sqrt{2m_n^*(E - E_C)}}{\pi^2 \hbar^3} dE \propto (E - E_C)^{1/2}$$
$$g_V(E)dE = \frac{m_p^* \sqrt{2m_p^*(E_V - E)}}{\pi^2 \hbar^3} dE \propto (E_V - E)^{1/2}$$

Density of States

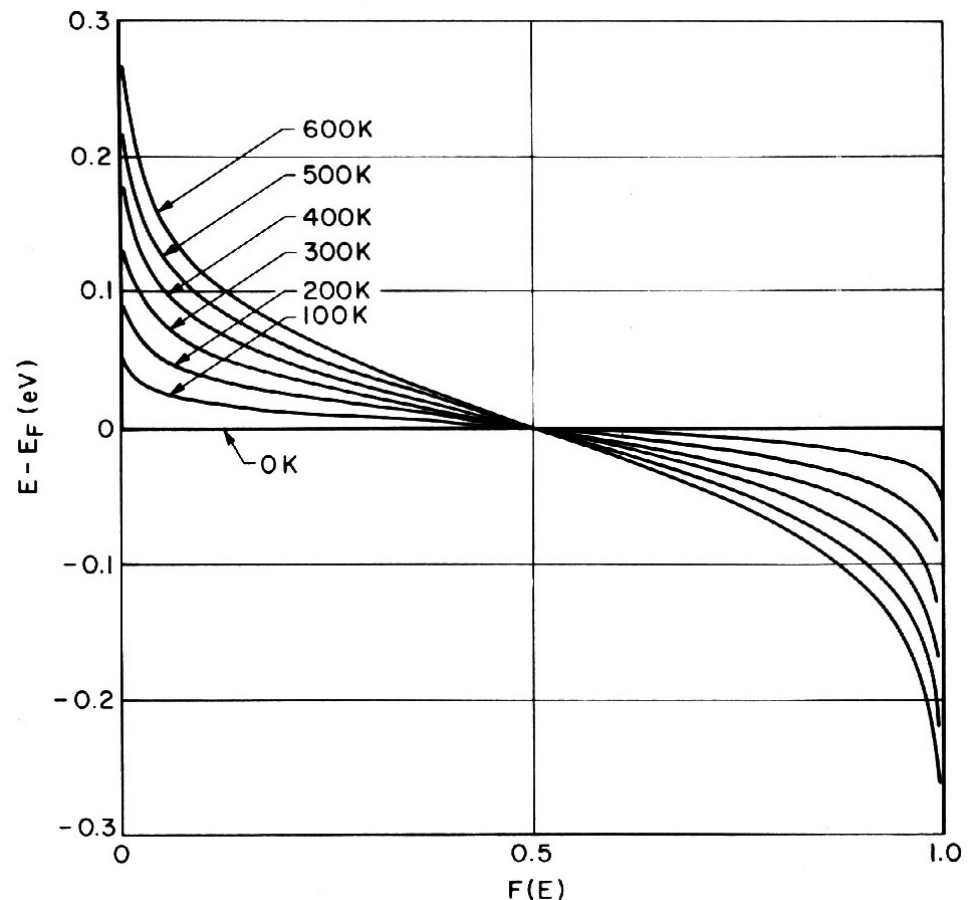


2.5.2 Fermi Function & Fermi Energy

- What determines whether an allowed state is occupied by an electron or not?
- Fermi Function $f(E)$:
 $f(E)$ is a probability function which gives the probability whether a state is occupied or not

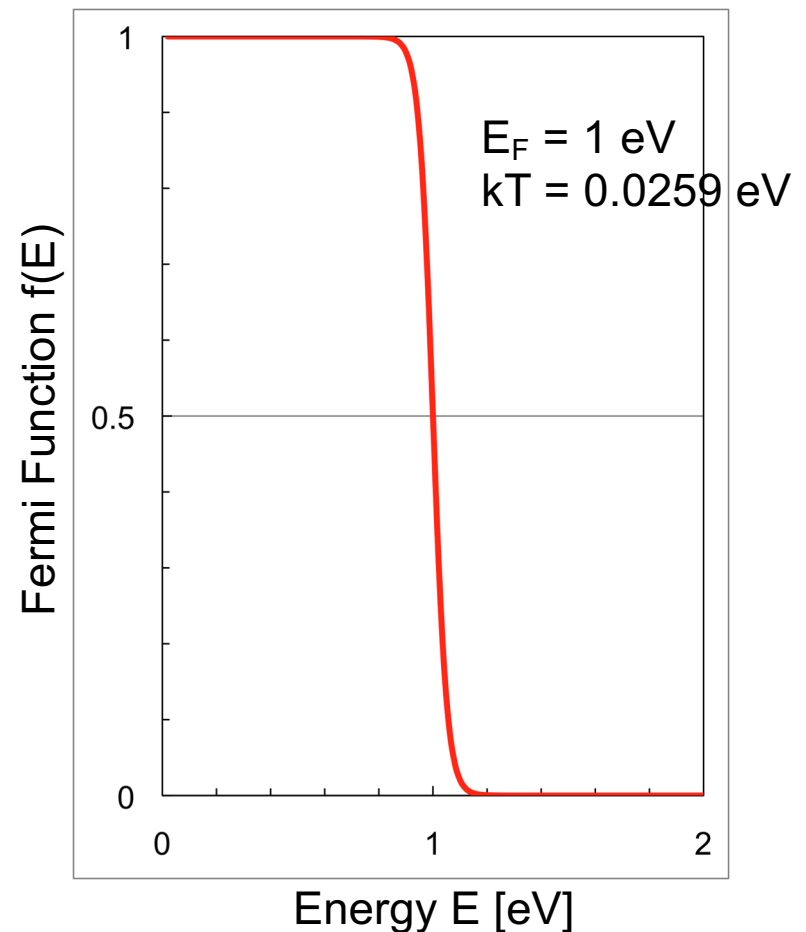
$$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}}$$

- E_F is the Fermi Energy



Characteristics of Fermi Function

- Because of Pauli principle:
 $0 \leq f(E) \leq 1$
- $f(E = E_F) = 0.5$
The probability that a state is occupied at the Fermi energy is 50%
- $f(E)$ is symmetric around E_F :
 $f(E_F + E) = 1 - f(E_F - E)$
- For $T = 0$ the Fermi function becomes a step function, i.e. all states below E_F are occupied, all states above E_F empty



Physical Interpretation of Fermi Energy E_F

- The Fermi energy E_F has the function of a thermodynamic potential
- **In thermodynamic equilibrium, E_F is constant across the device!!!**
- A gradient in E_F indicates non-equilibrium, resulting in a net flow of carriers

Boltzmann/Classical Approximation

- Away from $E = E_F$, i.e. for $|E - E_F| > 3kT$, the Fermi function can be approximated by exponential functions:

$$\begin{aligned} (E - E_F) > 3kT &\Rightarrow f(E) \approx e^{-(E-E_F)/kT} \\ (E_F - E) > 3kT &\Rightarrow f(E) \approx 1 - e^{-(E_F-E)/kT} \end{aligned}$$

- This approximation is called the **Boltzmann** or **Classical approximation**

2.5.3 Carrier Density n and p

- The carrier densities of electrons n [cm^{-3}] and holes p [cm^{-3}] can be calculated from the density of states $g(E)$ and the Fermi function $f(E)$:

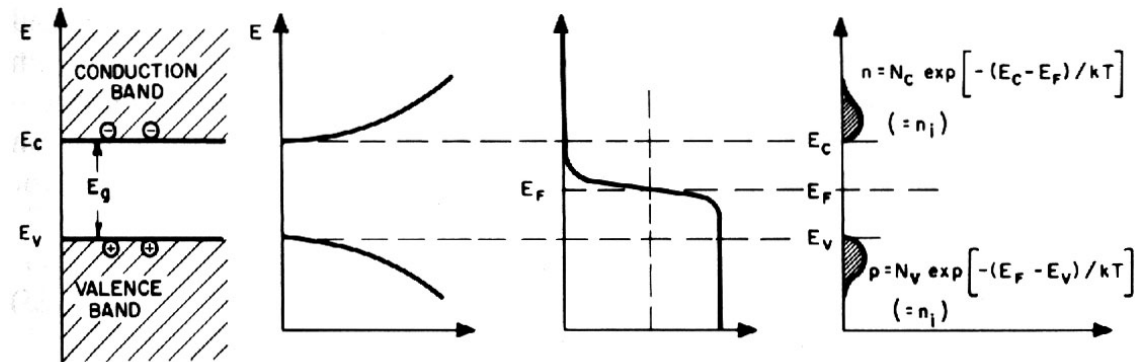
$$n = \int_{E_C}^{\text{Top} \rightarrow \infty} g_C(E) f(E) dE$$

$$p = \int_{\text{Bottom} \leftarrow -\infty}^{E_V} g_V(E) \underbrace{(1 - f(E))}_{\text{a hole is a non-occupied electron state}} dE$$

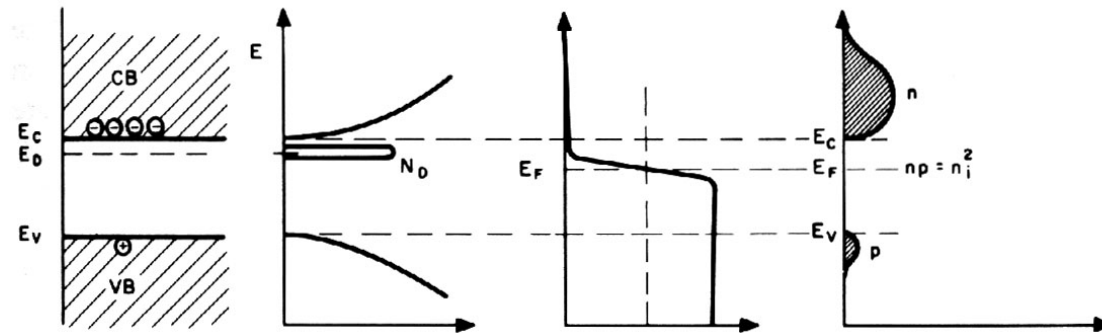
- The solution of the integral generally requires numerical methods; in case of the Boltzmann approximation however, the integrals can be solved analytically!

Carrier Concentration

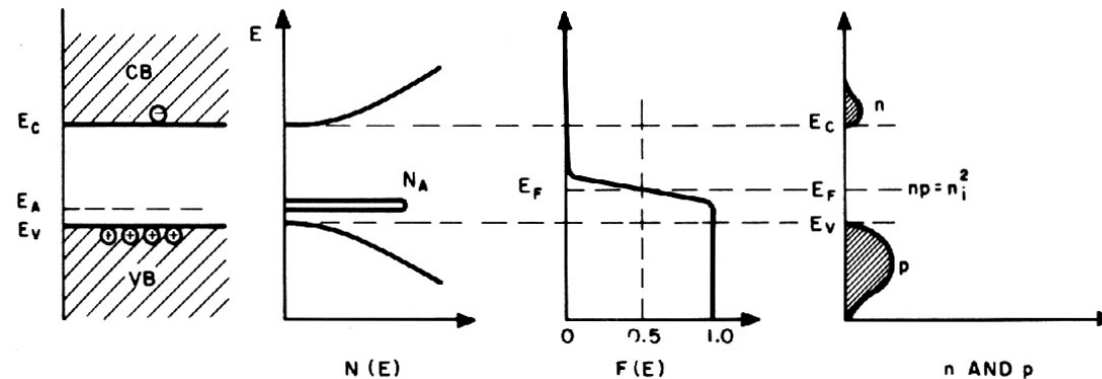
Intrinsic



Extrinsic N-Type



Extrinsic P-Type

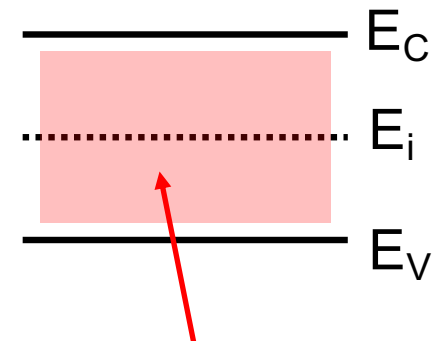


Carrier Density – Boltzmann Approximation

- For **electrons**: $(E_C - E_F) \geq 3kT$

$$n = N_C e^{(E_F - E_C)/kT}$$

$$N_C = 2 \left[\frac{m_n^* kT}{2\pi\hbar^2} \right]^{3/2} \approx \underbrace{2.86 \cdot 10^{19} \text{ cm}^{-3}}_{\text{Si @ 300K}}$$



$$E_C - 3kT \geq E_F \geq E_V + 3kT$$

- For **holes**: $(E_F - E_V) \geq 3kT$

$$p = N_V e^{(E_V - E_F)/kT}$$

$$N_V = 2 \left[\frac{m_p^* kT}{2\pi\hbar^2} \right]^{3/2} \approx \underbrace{2.66 \cdot 10^{19} \text{ cm}^{-3}}_{\text{Si @ 300K}}$$

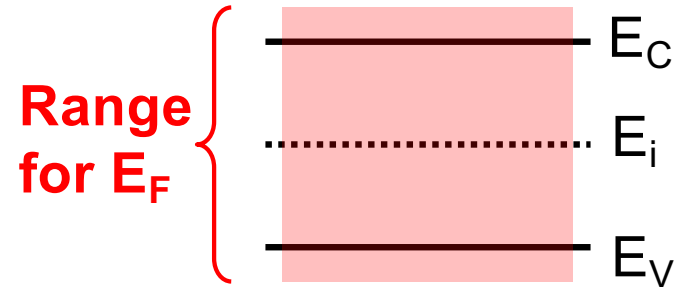
From Pierret:

$$N_{C,V} = 2.51 \cdot 10^{19} \left[\frac{m_{n,p}^*}{m_0} \right]^{3/2} \text{ cm}^{-3}$$

$$N_C \approx 3.2 \cdot 10^{19} \text{ cm}^{-3}$$

$$N_V \approx 1.8 \cdot 10^{19} \text{ cm}^{-3} \quad \text{for Si @ 300K}$$

Carrier Density – General Solution



- For **electrons**:

$$n = N_C \frac{2}{\sqrt{\pi}} F_{1/2}(\eta_C) \quad \text{with} \quad \eta_C = (E_F - E_C)/kT$$

- For **holes**:

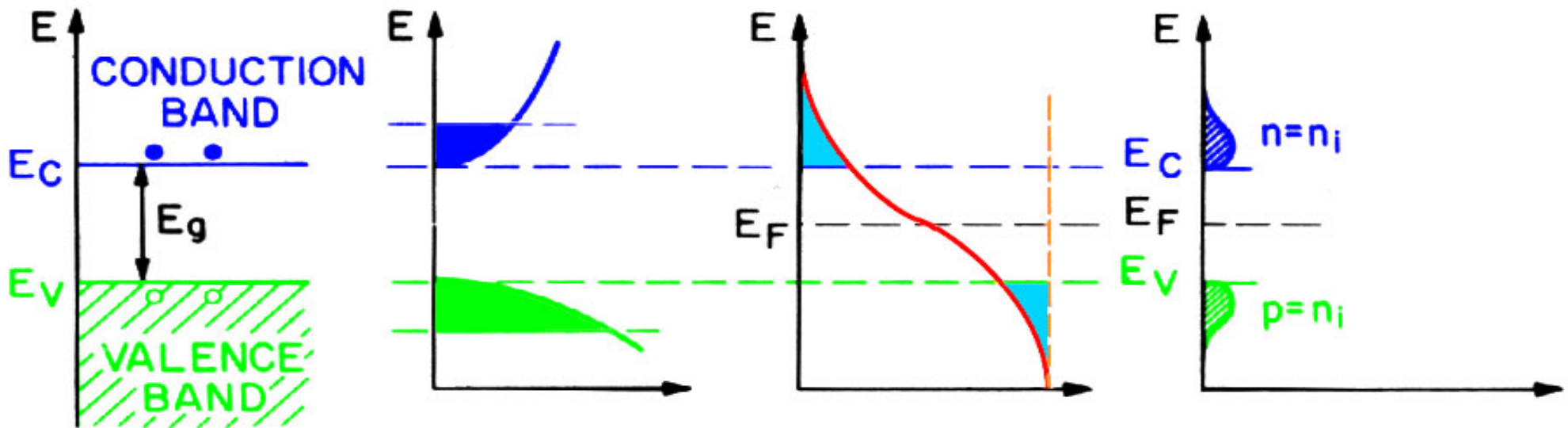
$$p = N_V \frac{2}{\sqrt{\pi}} F_{1/2}(\eta_V) \quad \text{with} \quad \eta_V = (E_V - E_F)/kT$$

- $F_{1/2}(\eta_C)$ is the **tabulated Fermi-Dirac integral** of order 1/2:

$$F_{1/2}(\eta_C) = \int_0^{\infty} \frac{\eta^{1/2}}{1 + e^{\eta - \eta_C}} d\eta$$

Intrinsic Semiconductor

- **No doping**
- Carrier concentration $n = p = n_i$ (for every electron in the conduction band, we have a hole, i.e. missing electron, in the valence band)
- Fermi energy $E_F = E_i$ close to center of band gap



Intrinsic Semiconductor

- **No doping**, i.e. intrinsic semiconductor with **$n = p = n_i$** and **$E_F = E_i$**
- From Boltzmann approximation with $E_F = E_i$, we obtain

$$n = N_C e^{(E_i - E_C)/kT} = n_i$$

$$p = N_V e^{(E_V - E_i)/kT} = n_i$$

- Thus, solving for n_i yields

$$n p = n_i^2 = N_C N_V e^{(E_V - E_C)/kT}$$

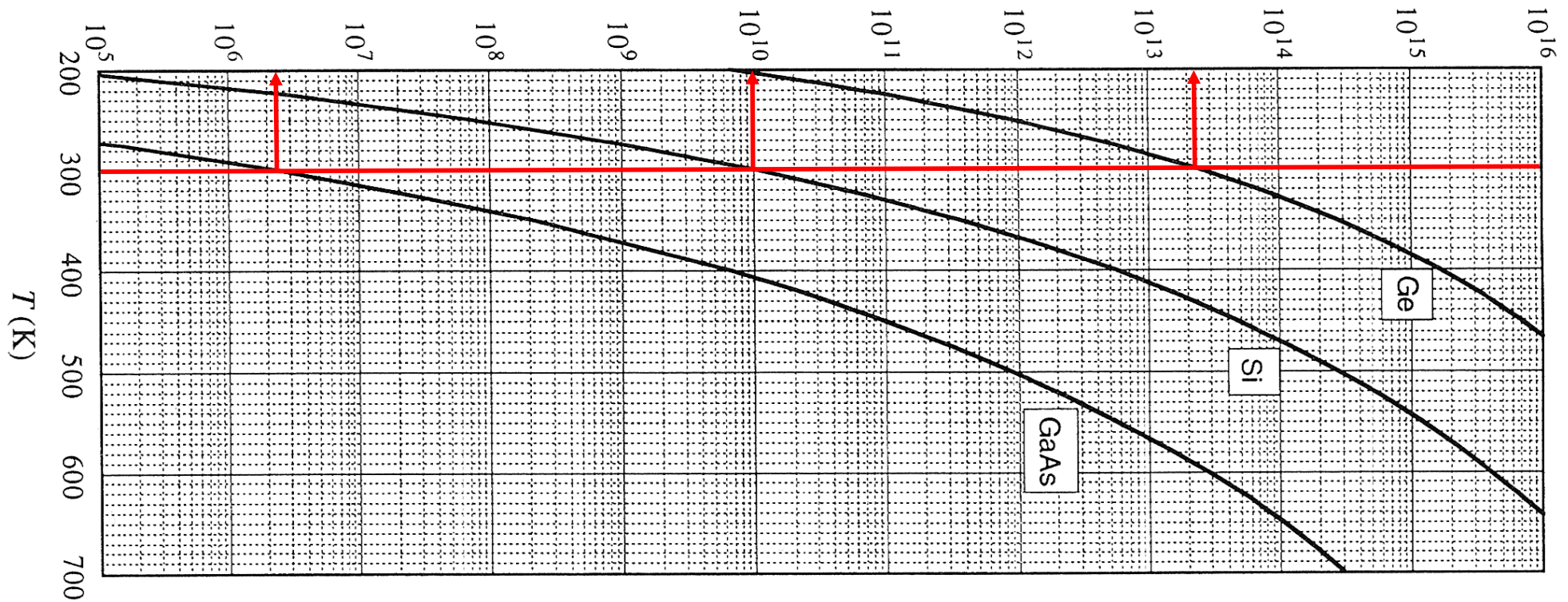
$$= N_C N_V e^{-E_g/kT}$$

$$n_i = \sqrt{N_C N_V} e^{-E_g/2kT} \approx \underbrace{10^{10} \text{ cm}^{-3}}_{\text{Si @ 300K}}$$

Intrinsic Semiconductor

– Carrier Concentration $n_i(T)$ –

Intrinsic carrier concentration (cm^{-3})



GaAs: $n_i(300\text{K}) = 2.25 \times 10^6 \text{ cm}^{-3}$

Ge: $n_i(300\text{K}) = 2.5 \times 10^{13} \text{ cm}^{-3}$

Si: $n_i(300\text{K}) = 1.0 \times 10^{10} \text{ cm}^{-3}$

Pierret, Fig. 2.20

Location of E_i

From the intrinsic carrier concentrations

$$p = n_i = N_V e^{(E_V - E_i)/kT}$$

$$n = n_i = N_C e^{(E_i - E_C)/kT}$$

With $n = p$, the intrinsic Fermi level E_i can be extracted

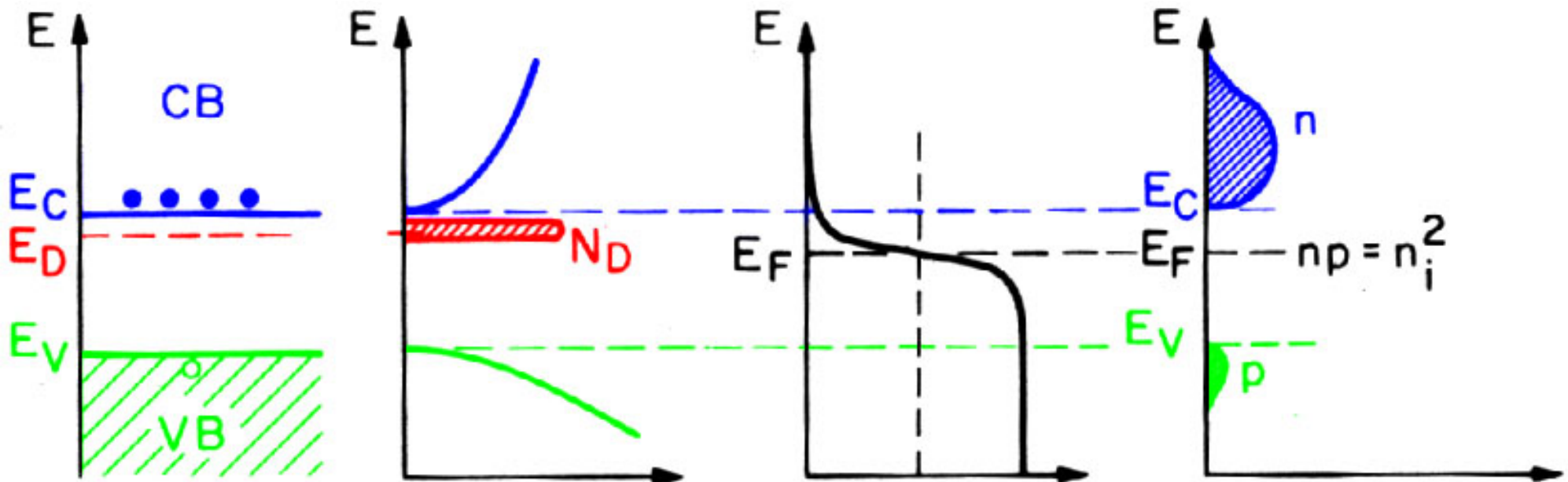
$$N_V e^{(E_V - E_i)/kT} = N_C e^{(E_i - E_C)/kT}$$

$$\Rightarrow E_i = \frac{E_C + E_V}{2} + \frac{3}{4}kT \ln \left[\frac{m_p^*}{m_n^*} \right]$$

The intrinsic Fermi level is only in the center of the band gap if $m_n^* = m_p^*$!!

Extrinsic Semiconductor

- **With doping**
- Carrier concentration $n \neq p \neq n_i$
- Example: n-type semiconductor



Extrinsic Semiconductor

- From Boltzmann approximation

$$n = N_C e^{(E_F - E_C)/kT} = \underbrace{N_C e^{(E_i - E_C)/kT}}_{=n_i} e^{(E_F - E_i)/kT}$$

$$n = n_i e^{(E_F - E_i)/kT}$$

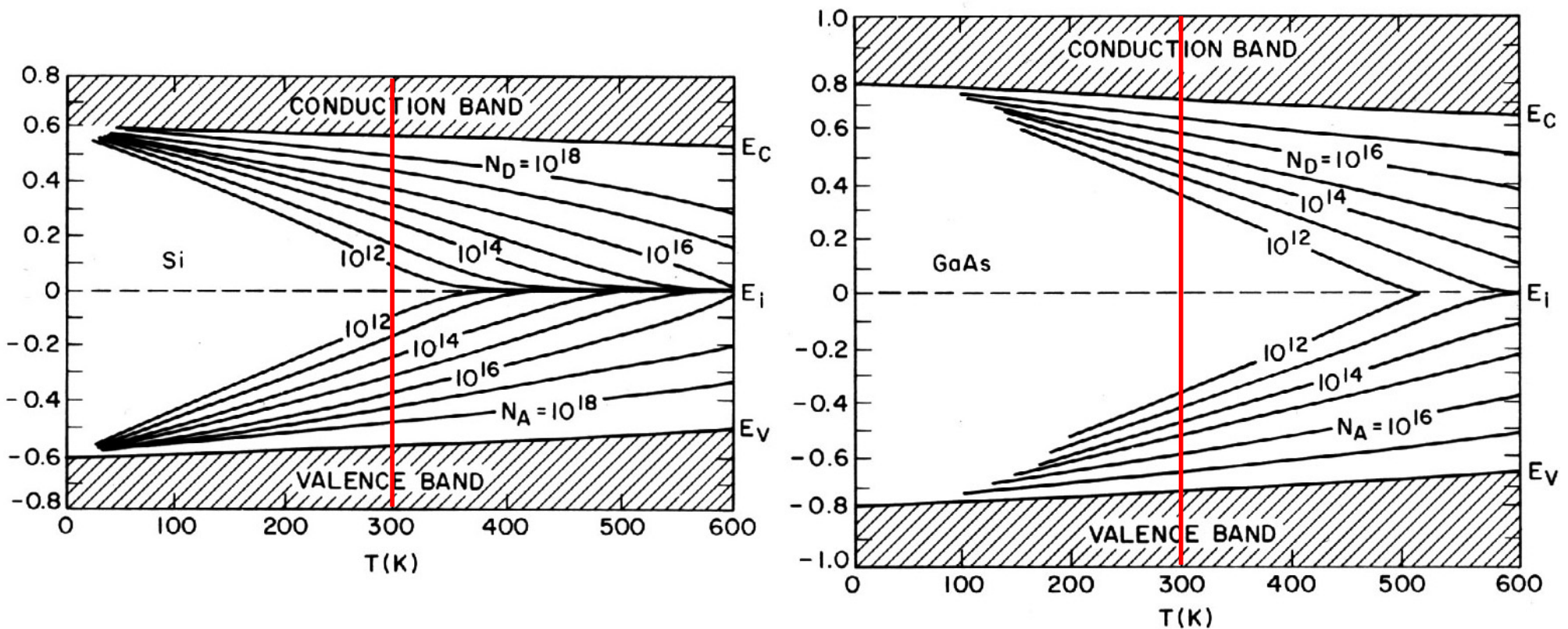
- Analogous for the holes

$$p = n_i e^{(E_i - E_F)/kT}$$

$$n p = n_i^2$$

Mass-Action-Law

Fermi Energy $E_F - E_i$ [eV] as a Function of Doping and Temp.



Sze, Fig. 2.28

2.5.4 Charge Neutrality Relation

- In equilibrium, **charge neutrality** is fulfilled:

$$\underbrace{p}_{\text{holes}} - \underbrace{n}_{\text{electrons}} + \underbrace{N_D^+}_{\text{ionized donors}} - \underbrace{N_A^-}_{\text{ionized acceptors}} = 0$$

- Assuming complete ionization $N_A^- = N_A$ and $N_D^+ = N_D$:

$$p - n + N_D - N_A = 0$$
$$np = n_i^2$$

Electron and Hole Concentration

- From mass-action-law and charge neutrality:

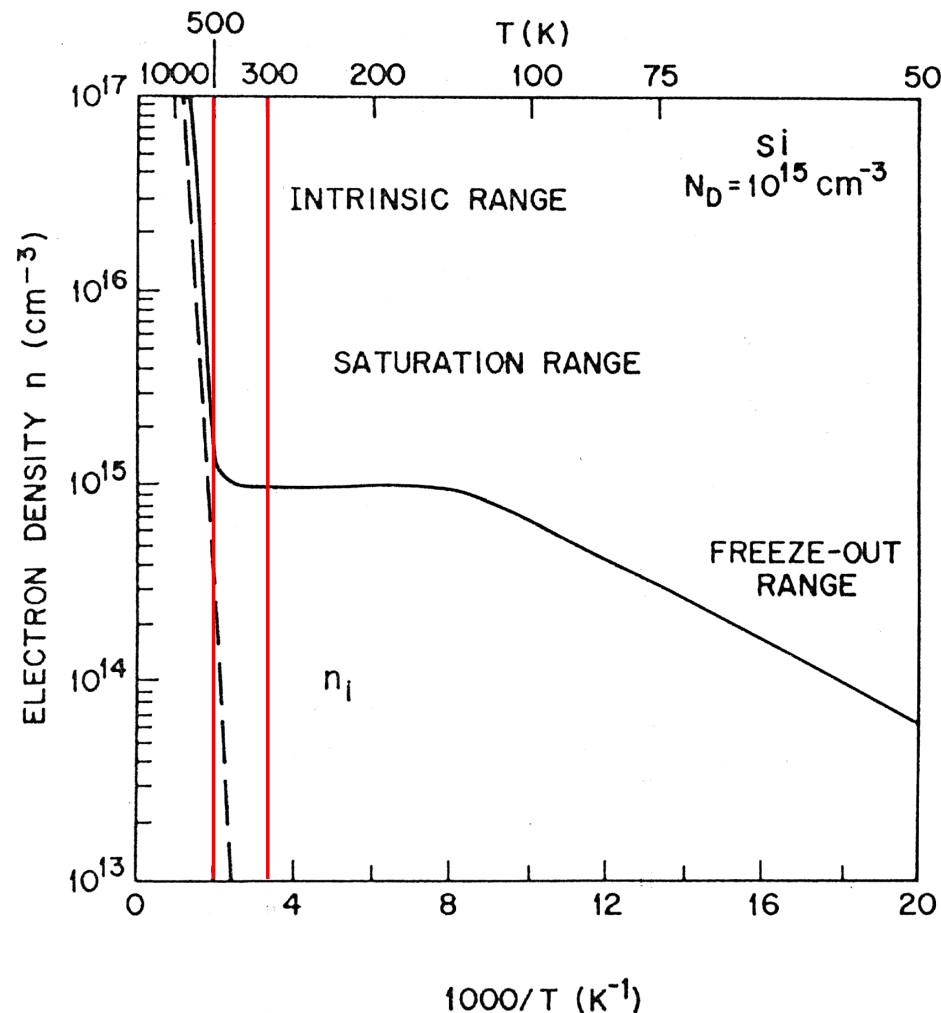
$$n = \frac{N_D - N_A}{2} + \sqrt{\left[\frac{N_D - N_A}{2}\right]^2 + n_i^2}$$
$$p = \frac{N_A - N_D}{2} + \sqrt{\left[\frac{N_A - N_D}{2}\right]^2 + n_i^2}$$

- Special Cases:

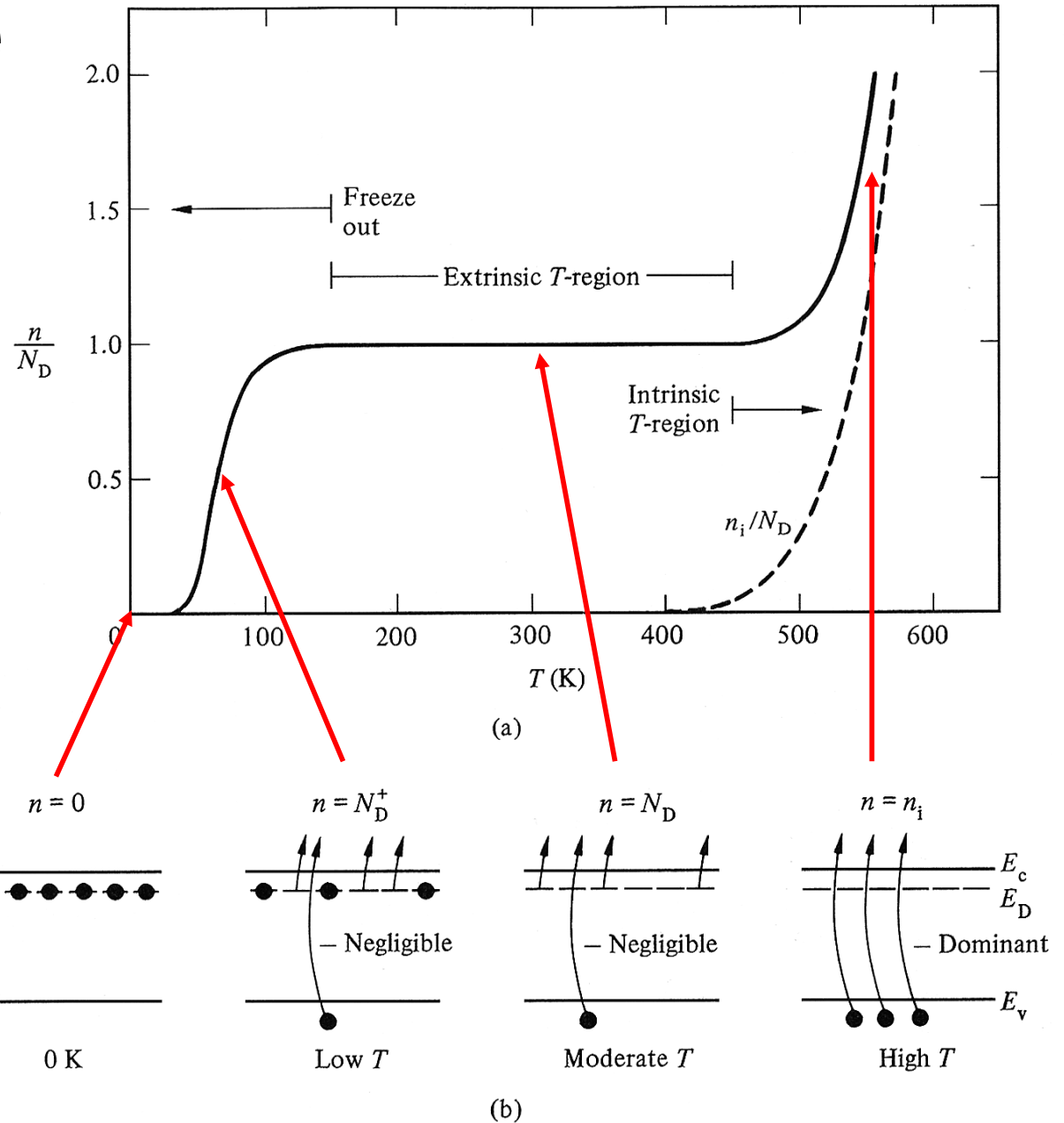
$$N_D \gg N_A \quad \text{and} \quad N_D \gg n_i : \quad n = N_D$$
$$p = n_i^2 / N_D$$

$$N_A \gg N_D \quad \text{and} \quad N_A \gg n_i : \quad p = N_A$$
$$n = n_i^2 / N_A$$

2.5.5 Temperature Dependence of Carrier Density



Temperature Dependence of Carrier Density



Pierret, Fig. 2.22

Carrier Statistics – Formula Summary

- Charge Neutrality & Mass-Action Law:
- Intrinsic Semiconductor
- Extrinsic Semiconductor
 - Linking Doping and Carrier Concentration
 - Linking Band Structure and Carrier Concentration